

Niteesh Krishna Penta

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Experience

Accenture Aug 2023 – Present
Software Engineer Hyderabad, India

Enterprise AI Chatbot for Employee Support Aug 2024 – Present

- Architected an enterprise-grade **Generative AI / RAG** chatbot across 7 microservices, processing large-scale unstructured data (PDFs, documents) for **entity extraction, summarisation, and semantic retrieval** using **LangChain**, **BAAI/bge-large-en-v1.5**, and **FAISS** vector search to deliver hallucination-free HR and IT support for **1,000+ employees**.
- Designed **agentic AI workflows** using **LangGraph** for multi-step reasoning and decision flows; built **Python-based APIs** to serve queryable AI insights, enabling downstream applications to consume analytical results with **safety guardrails** and **AI compliance** standards.
- Engineered **semantic response caching** using **BAAI** vector similarity in **Redis** (cosine similarity threshold 0.92) across **5 caching layers**, eliminating **60%** of LLM API calls and reducing latency by **25%**; integrated **Langfuse** for **LLM observability**, model evaluation, and production monitoring; applied **prompt engineering** to improve accuracy by **40%**.
- Developed **eval test suites** for model correctness using **RAGAS** metrics (faithfulness, context recall, answer relevancy); enforced **JWT** security and implemented input/output **guardrails** for responsible AI access across microservices.
- Deployed on **AWS EKS** with **Kubernetes** autoscaling, **Docker** containerisation, **CI/CD** via **GitHub Actions**, and observability via **Grafana** and **Prometheus**; implemented **Apache Kafka** event streaming with producer idempotence and manual offset acknowledgement.

IoT-based Solar Plant Monitoring System Aug 2023 – July 2024

- Developed scalable **RESTful APIs** using **Java Spring Boot** with **JWT** authentication and **Spring Security** for role-based access control; followed **TDD** reducing post-deployment bugs by **30%**.
- Designed **data ingestion and preprocessing pipelines** for **8,000+ daily IoT records**, enabling analytics-ready datasets for downstream processing; applied **statistical analysis** and **anomaly detection** algorithms on telemetry data stored in **PostgreSQL** with optimised queries cutting response time by **35%**.
- Built a **React** dashboard for real-time metrics visualisation via **WebSocket** for **200+ active users**; deployed on **AWS EC2** with **NGINX**, **Docker**, and **GitLab CI/CD** achieving **99.9% uptime**.

Education

National Institute of Technology Tiruchirappalli July 2019 – May 2023
B.Tech. in Mechanical Engineering with Minor in Computer Science CGPA: 8.44

Certifications

Microsoft Azure Fundamentals (AZ-900) – Microsoft June 2024
Microsoft Azure Data Engineer Associate (DP-203) – Microsoft March 2025

Skills

Languages: Python, Java/J2EE, SQL

AI & GenAI: RAG, LangChain, LangGraph, FAISS, BAAI/bge-large-en-v1.5, OpenAI API, Prompt Engineering, Agentic AI, Guardrails, NLP, Unstructured Data Processing, Vector Databases

MLOps & Evaluation: Langfuse, RAGAS, Model Evaluation Metrics, Anomaly Detection, Statistical Analysis, Observability Pipelines

Frameworks: FastAPI, Spring Boot, Spring MVC, Hibernate, Spring Security, React

Data & Messaging: PostgreSQL, Redis (NoSQL), Apache Kafka, Apache Spark (basics), ETL Pipelines, MySQL

Cloud & DevOps: AWS, Azure, Kubernetes, Docker, CI/CD (GitLab, GitHub Actions), Grafana, Git